DREADNOUGHT EXPLORATION PTY LTD

ANNUAL REPORT
MANGAROON REE PROJECT

For the Period

1 January 2022 to 31 December 2022

C253/2022 Annual Technical Report

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DISTRIBUTION:	

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Attachment 1	**Surface Sampling Attachments**
	20230331_E09_2620_2531.zip
	Historical Sampling.zip

ATTACHMENTS SUBMITTED SEPARATELY

1. Bibliographic Data Sheet

Project Name:	Mangaroon REE Project
Combined Reporting Number:	C253/2022
Tenement Numbers:	E 09/2620,E 09/2531
Tenement Operator(s):	DREADNOUGHT EXPLORATION PTY LTD
Report Type:	Annual
Report Title:	C253/2022 Annual Technical Report
Report Period:	1 January 2022 to 31 December 2022
Author:	Luke BLAIS
Submitted By:	Luke BLAIS
Report Date:	1 April 2023

Map Sheets:	*1:250,000 Map Sheet*	*1:100,000 Map Sheet*
	SF50-14 (EDMUND)	2150 (EDMUND)
	SF50-14 (EDMUND)	2250 (ELLIOTT CREEK)
	SG50-02 (MOUNT PHILLIPS)	2249 (MOUNT AUGUSTUS)
	SG50-02 (MOUNT PHILLIPS)	2149 (MOUNT PHILLIPS)

Target Commodity:	RARE EARTH ELEMENTS
Prospects Drilled:	
PoW Number:	
Geophysical Survey Reg No:	
Assays:	

Abstract

Location: Dreadnought Resources' Mangaroon project is in the Gascoyne Region, approximately 250kms south-east of Exmouth, near the junction of the shires of the Upper Gascoyne, Carnarvon, and Ashburton in the Gascoyne Region of Western Australia.

The tenements in this report are ~30km NW of Mt Augustus.

Geology: The project is across the 1:250,000 GSWA Edmund and Mount Phillips map sheets, and spans across the Edmund, Elliot Creek, Mount Augustus and Mount Phillips 1:100,000 maps. The project is mostly underline by Edmund Group and Durlacher Supersuite within the Gascoyne Province of the Capricorn Orogen. The tenements are located adjacent to the Yangibana and Yin REE projects to the WNW, and are north of the historical Cobra-Bangemall gold fields.

Work Done: Exploration in the reporting period comprised a data review, stream sediment sampling and lag sampling as part of early, first-pass exploration focusing on REE mineralisation in a potentially previously-overlooked area near the Gifford Creek Carbonatite Complex.

Results: No anomalies were identified in the stream sediment samples. There were some anomalous REE rock chips (up to 0.24% TREO) associated with a type of sedimentary conglomerate. There is one anomalous lag sample which carried 126ppm Ce.

Conclusion: Further work is required to understand the cause for the REE anomalism in the sedimentary conglomerate unit. Follow up lag sampling to check repeatability of the single point anomaly may be warranted.

Surface Geochemistry Summary:

Sample Type	No. of Samples
Lag	3
Stream Sediment	45
Rock Chip	7

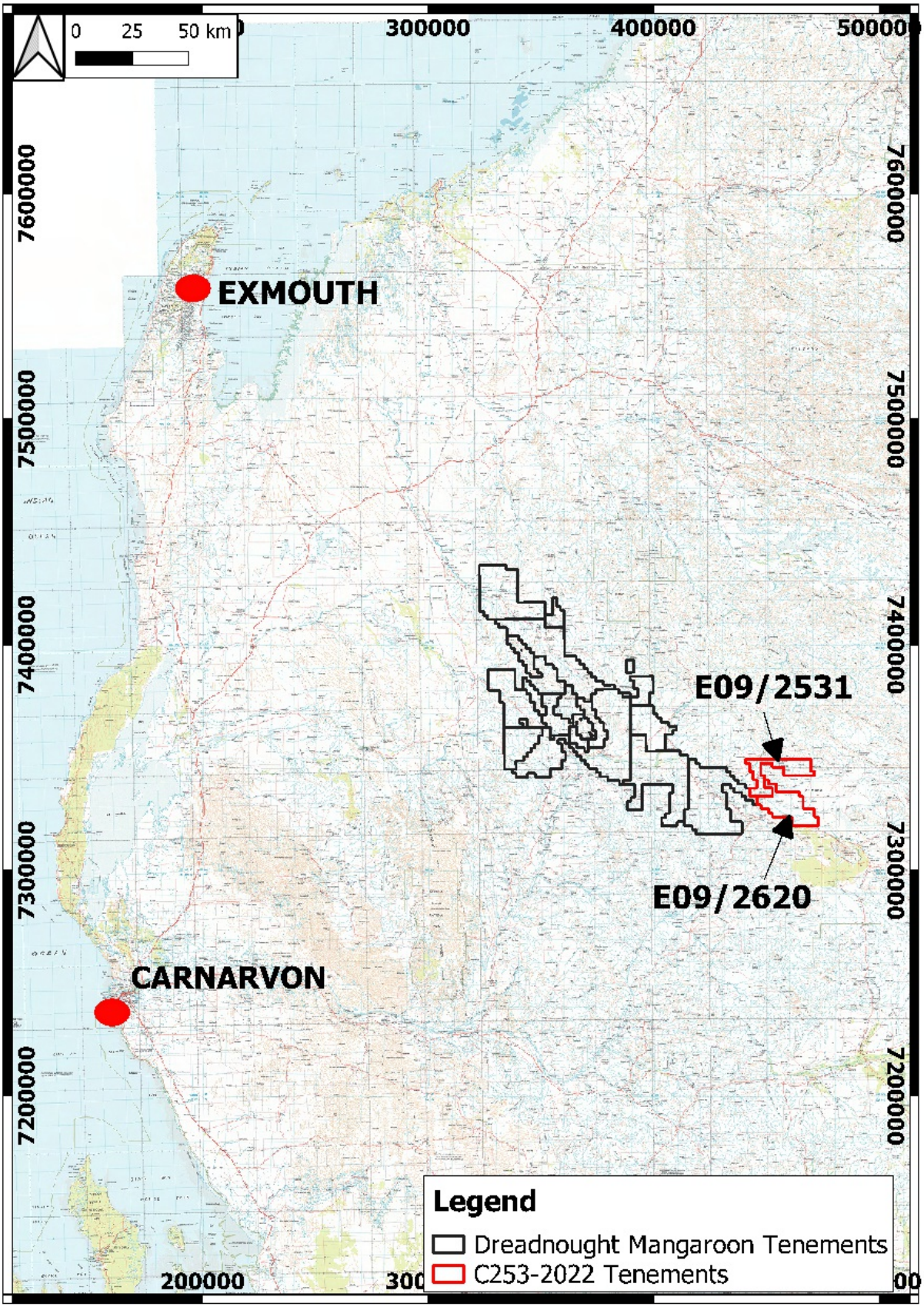
2. Introduction

The Combined Annual Report C253/2002 details work carried out on E09/2531 and E09/2620 which form part of Dreadnought Resources Mangaroon Project for the period 1/1/2022 to 31/12/2022.

The company is primarily exploring for REE mineralisation and acquired the ground due to the proximity to the Hastings Yangibana REE Project, and as part of the Dreadnought Yin REE Project. The tenements were of interest due to prominent thorium anomalies visible in state radiometric data.

3. Location and Access Details

The Mangaroon REE project is located ~250kms South-East of Exmouth, near the junction of the shires of Upper Gascoyne, Carnarvon, and Ashburton, in the Gascoyne Region of Western Australia.
The project can be accessed from Carnarvon/Gascoyne Junction via the Ullawarra/Gifford Creek road; from Exmouth via the Towera Road off the Great Northern Highway; from Meekatharra via Mount Augustas along national route 95.
The tenements in this combined report are approximately ~30km NW of Mt Augustus.

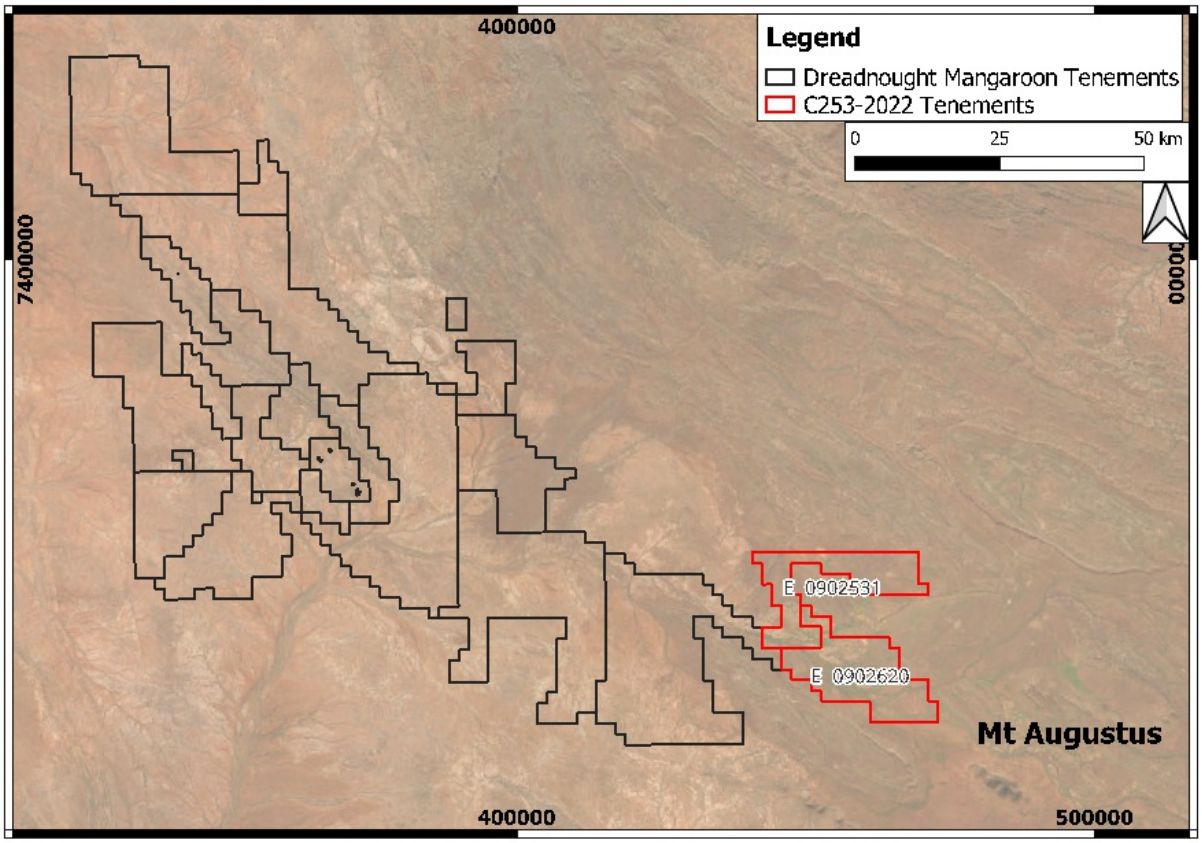


Location of C253/2022 Tenements in Mangaroon Project

4. Tenement Details

Tenement Information

Tenement	Grant Date	Expiry Date	Holder	Expenditure ($)	Area Size (KM2)	Area Size (BLK)
E 09/2531	07/02/2022	06/02/2027	DREADNOUGHT EXPLORATION PTY LTD	69000	193.2	69
E 09/2620	02/09/2022	01/09/2027	DREADNOUGHT EXPLORATION PTY LTD	82000	229.6	82



Tenement Plan for Mangaroon REE Project

5. Geology

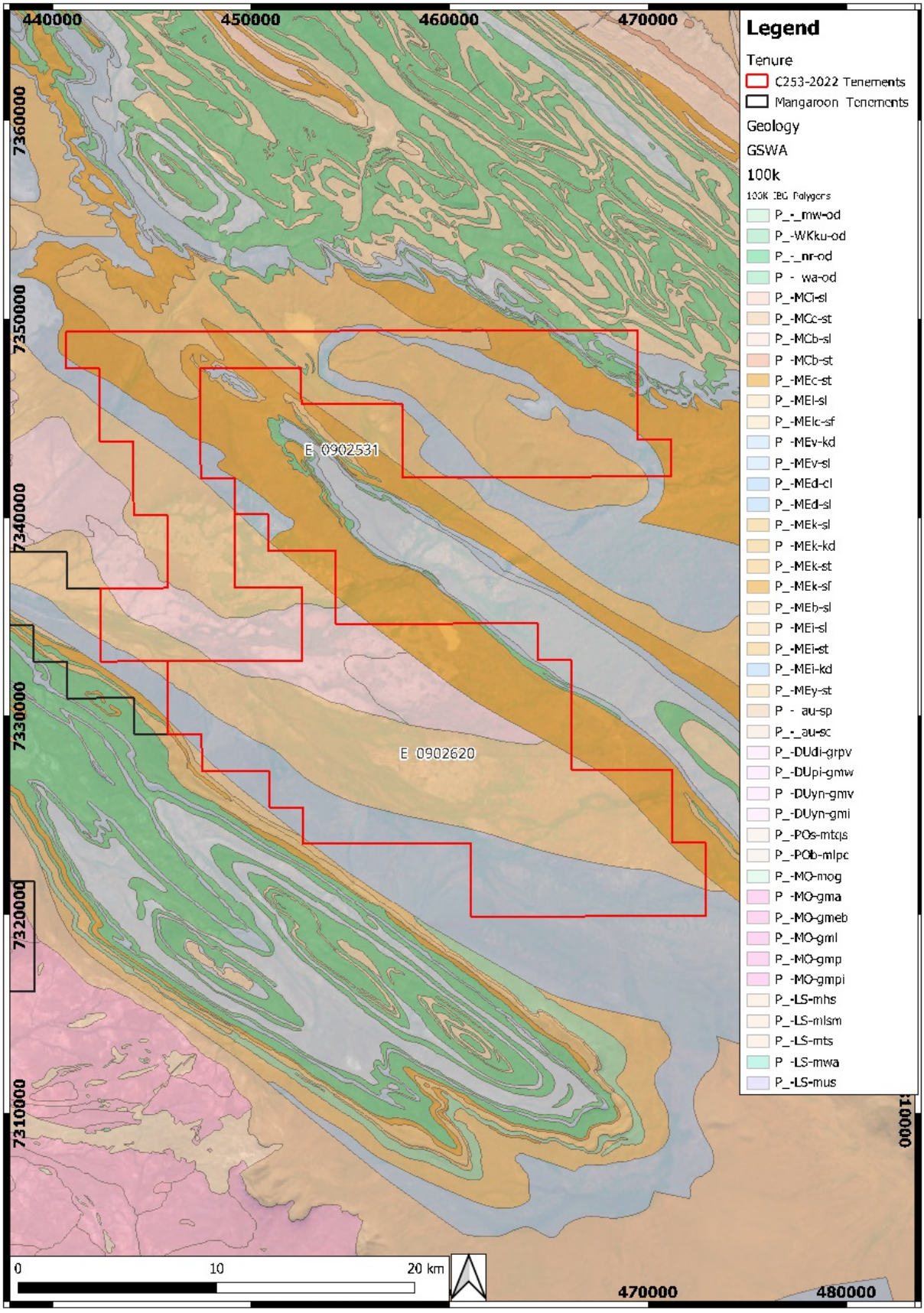
5.1 Regional Geology

The Mangaroon REE Project lies in the Gascoyne region of Western Australia between the Yilgarn Craton and the Pilbara Craton. The Gascoyne geological province consists of a basement of Neoarchean to Palaeoproterozoic granitic gneisses (the Glenburgh Terrane) overlain by various Proterozoic metasedimentary and granitic rocks (Johnson et. al 2011). The Glenburgh Terrane was accreted to the Pilabara Craton during the Ophthalmian Orogeny (c. 2215–2145 Ma). The combined Pilbara Craton and Glenburgh Terrane then collided with the Yilgarn craton during the Glenburgh Orogeny (c. 1950 Ma). Subsequent regional deformation occurred during the Capricorn Orogeny (c. 1820–1770 Ma) and Mangaroon Orogeny (c. 1680–1620 Ma) (Sheppard et. al 2010, Sheppard et. al 2016). The Gascoyne Province has recorded six tectono-thermal events between 2005–570 Ma and contains numerous crustal-scale faults and shear zones.

The Gascoyne Complex is fault-bounded to the east by younger Mesoproterozoic rocks belonging to the Edmund sedimentary basin which was affected by the Edmundian Orogeny reactivating faults and shear zones within the Gascoyne Complex as well as folding of sedimentary rocks of the Edmund Group. This Group together with the Collier Group comprises the Bangemall Supergroup. The Edmund Group is younger than 1680-1620 Ma granites in the underlying Gascoyne Complex and older than c. 1465 Ma dolerite sills that intrude it over large areas.

5.2 Local Geology

The project geology comprises Mesoproterozoic sedimentary rocks of the Edmund and Mount Augustus Basins. The Edmund Group in the project area is comprised of local fluvial to shallow marine sandstones and conglomerates (Mt Augustus Sandstone and Yilgatherra Formation) conformably overlain by more extensive dolostones including stromatolitic sections, variably pyritic dolomitic carbonaceous siltstones with interbeds of quartz sandstone forming the Irregully Formation. The Irregully Formation is overlain by the Kiangi Creek Formation consisting mainly of siltstones, mudstone, quartz sandstone and some turbidite.



Local geology of the C253-2022 report tenements

6. Previous Exploration

Geopeko (A12136) conducted magnetic, electromagnetic and gravity surveys in 1982/83 on two aeromagnetic anomalies which coincided with gossanous outcrop on the Centipede and Millipede prospects. A total of 20 vertical RAB holes totalling 724m were drilled on the Centipede prospect. The holes intersected granite, mica schist and immature quartz arenite. The bottom of 19 holes were analysed for Cu, Pb, Zn, Ag and Au. No anomalous results were returned. Three vertical RAB holes totalling 131m were drilled on the Millipede prospect. All the holes were terminated in alluvium. Geopeko concluded that the magnetic anomalies were sourced in the pre-Bangemall basement granite.

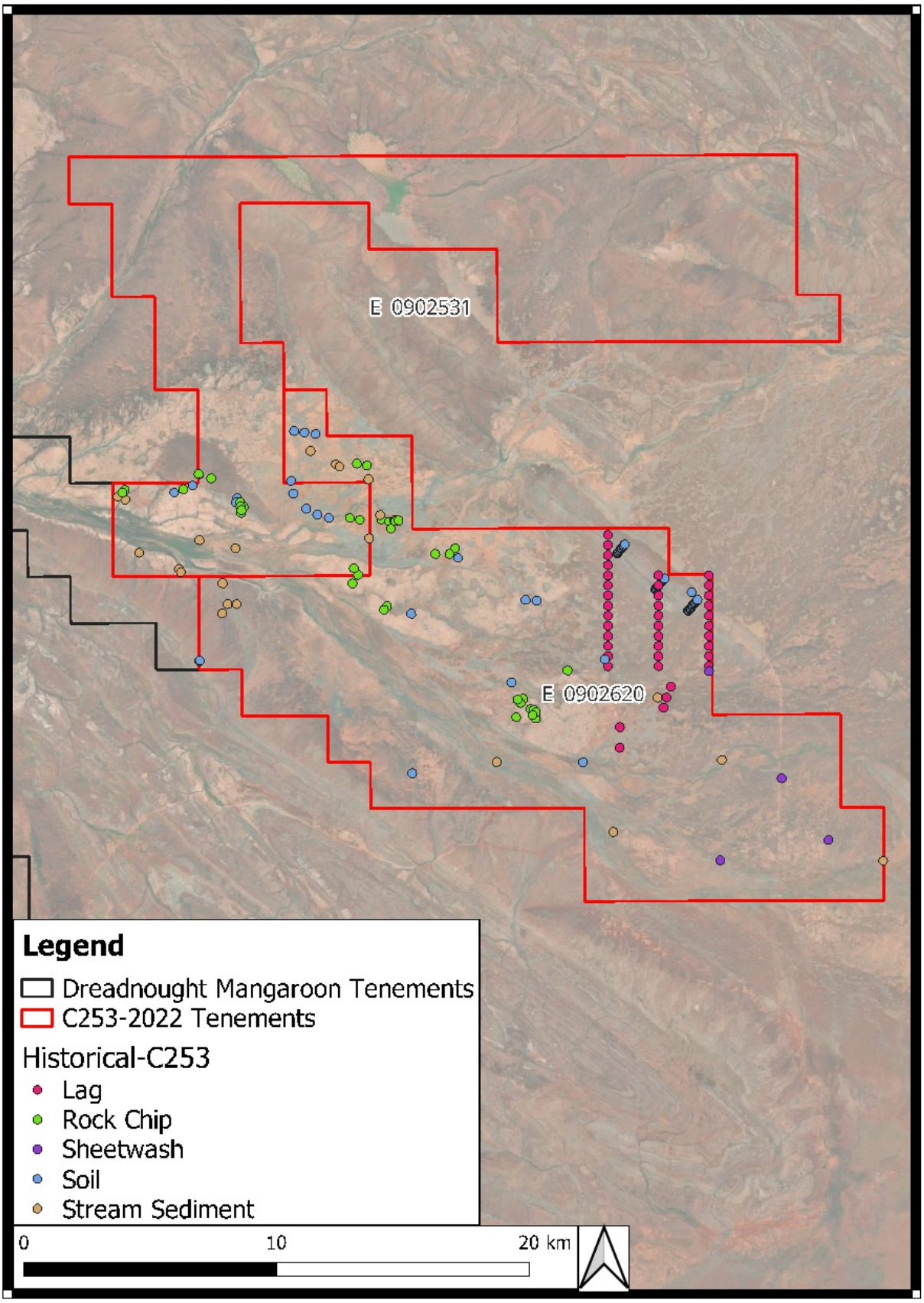
BHP took 2 soil samples in 1983-84 (A15960) which had limited results, and conclusions that the Cobra project had a low prospectivity for gold exploration. BHP Then took a further 37 Soil samples during 1985-1986 (A19281), targeting copper, lead zinc mineralisation in black shales for their Koorabooka project.

Newmont (A32886) took 8 Streams, 23 Soils and 45 Rock chips during the 1990-1991 period, as they were targeting base metals, gold and REE mineralisation.

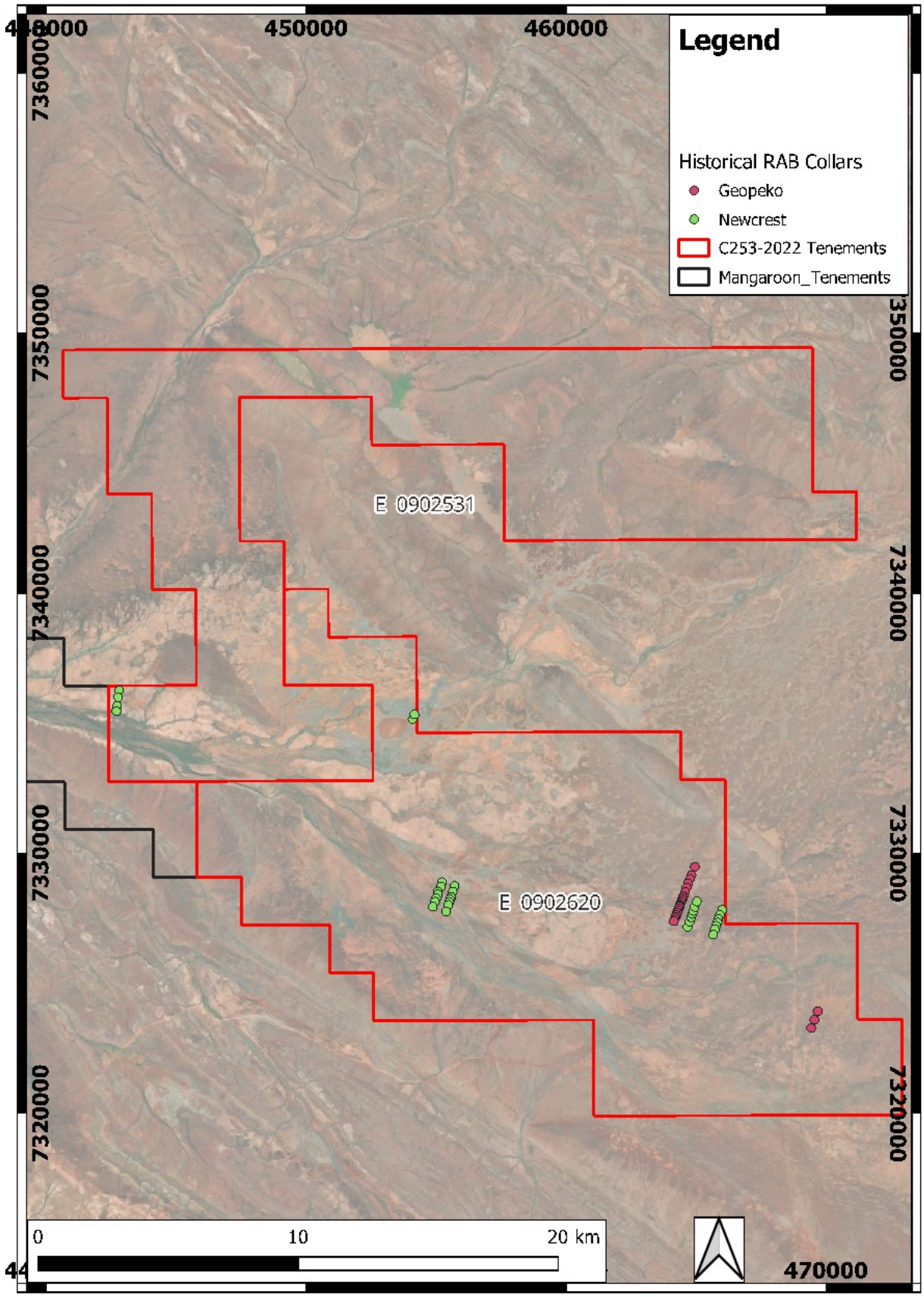
Newcrest (A34699) September 1991 drilled 34 RAB holes across four prospects; Centipede Bore, Jamieson Well, Quartz Hill and Spring Creek. No significant gold or basemetal results returned and the project was relinquished.

BHP (A51896) took 12 Stread Sediment samples during the 1996-1997 reporting period, targeting gold and base metal exploraiton around the historical Cobra mining centre.

Encounter Resources (A78776) completed gravity, ground water geochemistry and lag sampling. A total of 34 lag samples were collected over the tenement area targeting a WNW-trending magnetic feature. The lag samples were taken from an extensive broad sheet wash and the results do not appear to be related to the underlying sediments.



Historical Surface Samples from C253/2022 Tenement Area



Historical Drill Collars in C253/2022 Tenement Area

7. Current Exploration Summary

7.1 Data Review

Details of the data review are outlined in the historical exploration section.

Overall, historical exploration was centered around gold and base metal occurrences due to the proximity to the Cobra-Bangemall goldfields, as well as sharing similar stratigraphy to the Abra Pb-Ag-Cu-Au depoit. Due to recent discoveries by Dreadnought Resources adjacent to the Hastings Yangibana REE project, first-pass exploration targeting REE mineralisation, with appropriate assay methodology, was warranted.

7.2 Surface Sampling

Lag
Magnetic lag sampling was undertaken as part of first-pass exploration, and to test for REE anomalism incase subcropping ironstones (similar to those at Yangibana/Yin REE projects) were nearby, focusing on radiometric Th anomalies. 200g of magnetic lag material was collected into calico bags and analysed at Australian Laboratory Services (ALS) for ME-MS61L extended REE suite. Three lag samples were taken over the tenement area. One single point anomaly carrying a 126ppm Ce anomaly was returned, significantly higher than the other samples more in-line with background values (27 and 31ppm Ce).

Stream Sediment
45 Stream sediment samples were taken over the tenement area as part of a larger stream sediment program covering all Dreadnought Resources tenements at Mangaroon. The goal of the stream sediment program was to define either REE, base metal, or Au anomalism. Samples were taken at a nominal frequency of 1 sample per 5 km2. Material was collected from within the channel at each site from 1–3 locations 1–5 m apart. Banks and flood plains were avoided and samples were taken upstream from sources of contamination (road crossings, land slips, etc.) where possible.

No significant results were returned.

Rock Chip
>2kg of material was collected per rock chip sample, which were analysed at ALS for ME-MS61 and ME-XRF30

Four anomalous rock chip results (0.14-0.24% TREO) returned from a radioactive (Th) sedimentary conglomerate ridge. Maximum Th values up to 600ppm successfully explained the Th anomalism in the radiometrics, and were guided by sampling withe a scintillometre. It is unclear what was controlling the style of REE-Th mineralisation, whether the REE-Th resulted from hydrothermal activity or if it was in the primary sediments.



Example of rock chip that returned 0.15% TREO

8. Conclusion and Recommendations

Althought there was limited anomalism returned from the stream sediment sampling, further work is required to understand the cause for the RE-bearing metasedimentary conglomerates (0.15-0.24% TREO) that were associated with Th anomalies in government radiometrics. Follow up lag sampling to check repeatability of the single point Ce anomaly may be warranted.

9. Appendices

No Appendices as text are available